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Roll No

FT-7005(2)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Simulation and Process Modelling

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What is simulation? List a few advantages and disadvantages of simulation.
b) What are the state variable, input variable, out put variable, manipulated variable and disturbance variables? Explain with proper examples.
2. a) What do you mean by system modeling? Explain.
b) Distinguish between open and closed system with examples.
3. a) What is random variable? Explain all the types of random variables with suitable example.
b) Give the applications of the Poisson's random variables.
4. a) Explain Queuing theory with its all characteristics?
b) Differentiate between m/m/1 and m/m/c.
5. Compare and contrast validation scheme used experimental model and system dynamic models.

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6. a) Explain importance of mapping in case of uniform random variables.
b) Differentiate between linear and non-linear and stochastic models.
7. a) Explain distributed parameter model for packed bed.
b) Discuss the modeling of reactive separation processes.
8. a) Explain continuous and discrete models with suitable example.
b) Discuss the concept of continuous uniformly distributed Random number.
